

Practice Quiz 02B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Conditionals, contrapositives, and free and bound variables.

1. Let B mean *the battery charges* and P mean *the device is plugged in*. The statement is $B \rightarrow P$.

- (a) Write its converse and contrapositive as formulas. Make one four-row truth table and say which is equivalent to $B \rightarrow P$.

Solution The converse is $P \rightarrow B$ and the contrapositive is $\neg P \rightarrow \neg B$.

B	P	$B \rightarrow P$	$P \rightarrow B$	$\neg P \rightarrow \neg B$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	T	T	T

The contrapositive is equivalent to the original conditional; the converse is not. ■

- (b) In $\exists y \in \mathbf{Z} (y + m = n)$, list the free variables and the bound variables.

Solution The variable y is bound by $\exists y$. The variables m and n are free. ■